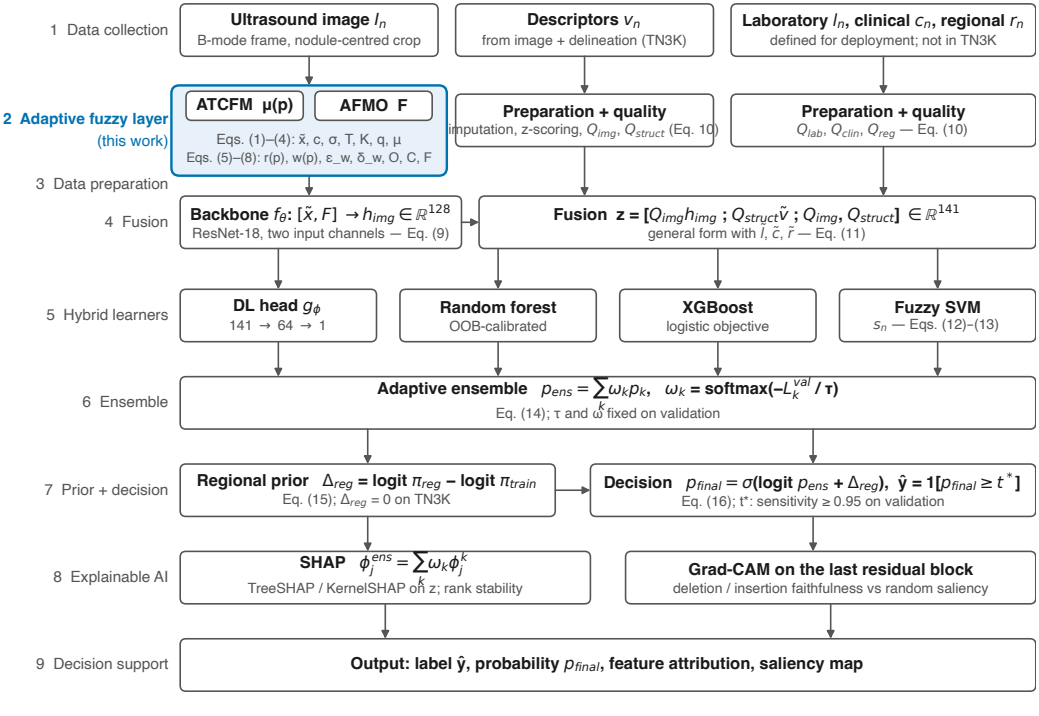

1 Data collection
Ultrasound image I_n
B-mode frame, nodule-centred crop
Descriptors v_n
from image + delineation (TN3K)
Laboratory l_n, clinical c_n, regional r_n
defined for deployment; not in TN3K
2 Adaptive fuzzy layer
(this work)
ATCFM μ(p)
AFMO F
Eqs. (1)–(4): x̃, c, σ, T, K, q, μ
Eqs. (5)–(8): r(p), w(p), ε_w, δ_w, O, C, F
Preparation + quality
imputation, z-scoring, Q_{img}, Q_{struct} (Eq. 10)
Preparation + quality
Q_{lab}, Q_{clin}, Q_{reg} — Eq. (10)
3 Data preparation
4 Fusion
Backbone f_θ: $[\tilde{x}, F] \to h_{img} \in \mathbb{R}^{128}$
ResNet-18, two input channels — Eq. (9)
Fusion $z = [Q_{img}h_{img}; Q_{struct}\tilde{v}; Q_{img}, Q_{struct}] \in \mathbb{R}^{141}$
general form with l, $\tilde{c}$, $\tilde{r}$ — Eq. (11)
5 Hybrid learners
DL head g_ϕ
141 → 64 → 1
Random forest
OOB-calibrated
XGBoost
logistic objective
Fuzzy SVM
s_n — Eqs. (12)–(13)
6 Ensemble
Adaptive ensemble $p_{ens} = \sum_k \omega_k p_k$, $\omega_k = \mathrm{softmax}(-L_k^{val} / \tau)$
Eq. (14); τ and ω fixed on validation
7 Prior + decision
Regional prior $\Delta_{reg} = \mathrm{logit}\,\pi_{reg} - \mathrm{logit}\,\pi_{train}$
Eq. (15); $\Delta_{reg} = 0$ on TN3K
Decision $p_{final} = \sigma(\mathrm{logit}\,p_{ens} + \Delta_{reg})$, $\hat{y} = \mathbf{1}[p_{final} \geq t^*]$
Eq. (16); t*: sensitivity ≥ 0.95 on validation
8 Explainable AI
SHAP $\phi_j^{ens} = \sum_k \omega_k \phi_j^k$
TreeSHAP / KernelSHAP on z; rank stability
Grad-CAM on the last residual block
deletion / insertion faithfulness vs random saliency
9 Decision support
Output: label ŷ, probability p_{final}, feature attribution, saliency map